

A Terminal-Lag Exponential-Tilt Phase Transition for the Complete Upper Counterloop Band

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Abstract

We determine the full exponential-transform phase diagram of the deterministic terminal-lag distribution isolated in RH-358. Let

$$\pi_k(r) = \frac{y_k^{2k-1-r}/(2k-1-r)}{C_k}, \quad 0 \leq r \leq k-2,$$

where $y_k = x \exp[-\log C_M/k + o(k^{-1})]$ and $x > 1$, and define $G_k(z) = \sum_r z^r \pi_k(r)$. For fixed $0 \leq z < x$,

$$G_k(z) \longrightarrow \frac{1-x^{-1}}{1-z/x}.$$

In the critical window $z_k = x \exp(\tau/k)$,

$$\frac{G_k(z_k)}{k} \longrightarrow 2(1-x^{-1}) \int_0^1 \frac{e^{(\tau+\log C_M)s}}{2-s} ds.$$

For fixed $z > x$,

$$G_k(z) \sim \frac{2C_M(1-x^{-1})}{1-x/z} \left(\frac{z}{x}\right)^{k-2}.$$

The tilted laws converge respectively to a terminal geometric distribution, a continuous critical density on the scaled lag r/k , and an opposite-endpoint geometric distribution. Thus $k^{-1} \log G_k(z)$ has the sharp limit $\max\{0, \log(z/x)\}$. These are normalized deterministic budget laws, not spectral or noisy stochastic distributions. Actual-head inheritance remains conditional on the original unnormalized same-clock leaf $D_{4k}(R) \rightarrow 0$. No determinant, Gate A–E, Hilbert–Polya, zero-identification, or RH claim is made.

1 Exact transform identity

The source-locked radial law is

$$y_k = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right], \quad x > 1, \quad C_M > 0. \quad (1)$$

RH-358 defines the complete deterministic strict upper budget

$$C_k = \sum_{r=0}^{k-2} \frac{y_k^{2k-1-r}}{2k-1-r} \quad (2)$$

and the terminal-lag probability

$$\pi_k(r) = \frac{y_k^{2k-1-r}/(2k-1-r)}{C_k}, \quad 0 \leq r \leq k-2, \quad (3)$$

extended by zero to $\mathbb{Z}_{\geq 0}$ [3]. It converges in total variation to the geometric law $(1-x^{-1})x^{-r}$. That convergence alone does not control arbitrary exponential observables; the present theorem finds the exact uniform-integrability boundary.

For $z \geq 0$, define

$$G_k(z) := \sum_{r=0}^{k-2} z^r \pi_k(r). \quad (4)$$

For $u \geq 0$, introduce the finite sum

$$A_k(u) := \sum_{r=0}^{k-2} u^r \frac{2k-1}{2k-1-r}. \quad (5)$$

Proposition 1.1 (Exact transform quotient). *For every $z \geq 0$,*

$$\boxed{G_k(z) = \frac{A_k(z/y_k)}{A_k(1/y_k)}}. \quad (6)$$

Moreover,

$$A_k(1/y_k) \longrightarrow \frac{1}{1-x^{-1}}. \quad (7)$$

Proof. Multiply numerator and denominator of (4) by $(2k-1)y_k^{-(2k-1)}$ to obtain (6). For each fixed r , $y_k^{-r}(2k-1)/(2k-1-r) \rightarrow x^{-r}$. Choose $1 < \xi < x$ with $y_k \geq \xi$ eventually. On the support, $(2k-1)/(2k-1-r) < 2$, so the summands are bounded by $2\xi^{-r}$. Dominated convergence gives (7). $\square$

2 Subcritical, critical, and supercritical transforms

Theorem 2.1 (Three transform regimes). *The following statements hold.*

1. *If $0 \leq z < x$ is fixed, then*

$$\boxed{G_k(z) \longrightarrow \frac{1-x^{-1}}{1-z/x}}. \quad (8)$$

The convergence is locally uniform on $[0, x)$.

2. *Fix $\tau \in \mathbb{R}$ and put*

$$z_k = x \exp(\tau/k), \quad \eta := \tau + \log C_M. \quad (9)$$

Then

$$\boxed{\frac{G_k(z_k)}{k} \longrightarrow \Phi(\eta) := 2(1-x^{-1}) \int_0^1 \frac{e^{\eta s}}{2-s} ds}. \quad (10)$$

3. *If $z > x$ is fixed, then*

$$\boxed{G_k(z) \sim \frac{2C_M(1-x^{-1})}{1-x/z} \left(\frac{z}{x}\right)^{k-2}}. \quad (11)$$

Proof. For the first statement, $z/y_k \rightarrow z/x < 1$. On any compact subinterval below x , the numerator in (6) is dominated by one common geometric sequence. Thus

$$A_k(z/y_k) \longrightarrow \frac{1}{1-z/x};$$

combine with (7).

In the critical window, (1) gives

$$k \log(z_k/y_k) \longrightarrow \eta. \quad (12)$$

Uniformly for $0 \leq r \leq k-2$,

$$(z_k/y_k)^r = e^{\eta r/k + o(1)}, \quad \frac{2k-1}{2k-1-r} = \frac{2}{2-r/k} + o(1).$$

Therefore the Riemann sum satisfies

$$\frac{1}{k} A_k(z_k/y_k) \longrightarrow 2 \int_0^1 \frac{e^{\eta s}}{2-s} ds.$$

Division by (7) proves (10).

For $z > x$, write $u_k = z/y_k \rightarrow z/x > 1$ and reverse the numerator sum:

$$A_k(u_k) = u_k^{k-2} \frac{2k-1}{k+1} \sum_{\ell=0}^{k-2} u_k^{-\ell} \frac{k+1}{k+1+\ell}. \quad (13)$$

Dominated convergence gives a sum limit $(1-x/z)^{-1}$, while the prefactor ratio tends to two. Finally,

$$u_k^{k-2} \sim C_M (z/x)^{k-2} \quad (14)$$

by (1). Use (7) to obtain (11). $\square$

Corollary 2.2 (Free-energy transition). *For every fixed $z \geq 0$,*

$$\boxed{\lim_{k \rightarrow \infty} \frac{1}{k} \log G_k(z) = \max \left\{ 0, \log \frac{z}{x} \right\}}, \quad (15)$$

where the right side is interpreted as zero at $z = 0$.

Proof. For $z < x$, (8) has a finite positive limit. At $z = x$, use (10) with $\tau = 0$ to get $G_k(x) \sim k\Phi(\log C_M)$. For $z > x$, take logarithms in (11). $\square$

Remark 2.3 (The multiplier drift is critical). The subcritical transform forgets C_M . At $z = x$, the leading constant is $\Phi(\log C_M)$, and above x the factor C_M appears explicitly. Replacing y_k by x before taking the critical or supercritical limit loses a leading constant.

3 The three tilted distribution limits

Whenever $G_k(z) > 0$, define the mathematical tilted probability

$$\pi_{k,z}(r) := \frac{z^r \pi_k(r)}{G_k(z)}. \quad (16)$$

Theorem 3.1 (Tilted phase diagram). *The tilted laws have the following limits.*

1. For fixed $0 \leq z < x$, extended by zero off the finite support,

$$\pi_{k,z} \longrightarrow (1-z/x)(z/x)^r \quad (17)$$

in total variation on $\mathbb{Z}_{\geq 0}$, with the $z = 0$ law interpreted as δ_0 .

2. For $z_k = x \exp(\tau/k)$, the scaled lag R_k/k under π_{k,z_k} converges weakly on $[0, 1]$ to the density

$$f_\eta(s) = \frac{e^{\eta s}/(2-s)}{\int_0^1 e^{\eta u}/(2-u) du}, \quad \eta = \tau + \log C_M. \quad (18)$$

3. For fixed $z > x$, let $L_k = k - 2 - R_k$. Then

$$\mathbb{P}_{\pi_{k,z}}(L_k = \ell) \longrightarrow (1 - x/z)(x/z)^\ell \quad (19)$$

in total variation on $\mathbb{Z}_{\geq 0}$.

Proof. The tilted unnormalized weights are

$$(z/y_k)^r \frac{2k-1}{2k-1-r}. \quad (20)$$

Below x , these weights converge pointwise to $(z/x)^r$ and have a common summable geometric envelope. Dominated convergence before and after normalization proves (17).

In the critical window, for every continuous g on $[0, 1]$, the same Riemann sum used in (10) gives

$$\frac{1}{k} \sum_{r=0}^{k-2} g(r/k) (z_k/y_k)^r \frac{2k-1}{2k-1-r} \longrightarrow 2 \int_0^1 g(s) \frac{e^{\eta s}}{2-s} ds.$$

Divide by the case $g = 1$ to obtain (18).

Above x , substitute $r = k - 2 - \ell$ into (20) and divide by the $\ell = 0$ weight. The ratio tends to $(x/z)^\ell$ and is dominated by a common geometric sequence. Normalization proves (19). $\square$

Remark 3.2 (No spectral probability claim). The probabilities in (16) are normalized deterministic absolute budget weights. They are not eigenvalue distributions, root-counting measures, Gibbs states of the noisy operator, or local probability laws for the dynamics.

4 Conditional actual-head inheritance

Let $\pi_k^{\mathcal{H}}$ be the actual even-weight lag distribution defined in RH-358. Under the original same-clock unnormalized leaf

$$D_{4k}(R) := \sum_{2 \leq n < 4k} \frac{|h_{\sigma,n} - s_{k,n}| R^n}{n} \longrightarrow 0, \quad (21)$$

RH-358 proves the stronger coordinatewise conclusion

$$\delta_k := \sup_{0 \leq r \leq k-2} \left| \frac{\pi_k^{\mathcal{H}}(r)}{\pi_k(r)} - 1 \right| \longrightarrow 0. \quad (22)$$

For any sequence $z_k \geq 0$, put

$$G_k^{\mathcal{H}}(z_k) := \sum_r z_k^r \pi_k^{\mathcal{H}}(r). \quad (23)$$

Theorem 4.1 (Uniform conditional tilt transfer). *Assume (21). For every nonnegative tilt sequence,*

$$\left| \frac{G_k^{\mathcal{H}}(z_k)}{G_k(z_k)} - 1 \right| \leq \delta_k \longrightarrow 0. \quad (24)$$

Moreover, if $z_k > 0$ and $\pi_{k,z_k}^{\mathcal{H}}$ is the normalized actual tilted law, then

$$\sup_r \left| \frac{\pi_{k,z_k}^{\mathcal{H}}(r)}{\pi_{k,z_k}(r)} - 1 \right| \longrightarrow 0. \quad (25)$$

If $z_k = 0$, both tilted laws are exactly δ_0 whenever they are defined. Consequently every transform and tilted-law conclusion above transfers only under (21).

Proof. The ratio in (24) is a weighted average of $\pi_k^{\mathcal{H}}(r)/\pi_k(r)$ with nonnegative weights $z_k^r \pi_k(r)$, so its distance from one is at most δ_k . Equation (25) follows by dividing the coordinatewise ratio by $G_k^{\mathcal{H}}(z_k)/G_k(z_k)$. $\square$

The hypothesis (21) remains open. This conditional theorem transfers budget laws only and contains no root, rank, or determinant identification.

5 Finite protocol, boundary, and review trigger

The artifact uses the exact rational fixture $x = 2352/1445$, $C_M = 1$ to reproduce exact subcritical and supercritical transforms, plus high-precision critical-window Riemann-integral diagnostics and tilted-law distances. Finite rows do not prove the limiting regimes.

The next paper is RH-361, the ten-layer review of RH-352–RH-361. It should audit the deterministic/actual boundary, preserve every open hypothesis, build the batch archive, and update the repository handoff.

RH-360 does not prove (21); identify actual roots, ranks, or spectra; close a determinant, p , the unrelated open q , or E_{off} ; close the moving noisy envelope of RH-241 [1]; or activate the gluing criterion of RH-288 [2]. Gates A–E remain false/open. There is no Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace theorem, completed-zeta divisor equality, or proof of RH.

References

- [1] Bin Wang. A ten-layer trace-envelope frontier review, 2026. Repository paper RH-241.
- [2] Bin Wang. A weighted prefix–tail gluing criterion for regularized determinants, 2026. Repository paper RH-288.
- [3] Bin Wang. Terminal-lag geometric localization for the complete upper counterloop band, 2026. Repository paper RH-358.